Question	Answer	Marks	Partial Marks
1(a)(i)	1 | 7 7 8 8 9 9 2 | 1 1 1 1 2 3 3 4 5	2	**B1** for one row correctly ordered or for fully correct unordered stem-and-leaf diagram or for a correct diagram with one error or omission
1(a)(ii)	21	1	
1(a)(iii)	23	1	
1(a)(iv)	48	2	**M1** for $\frac{2}{15}[\times 360]$ or $\frac{360}{15}[\times 2]$
1(b)(i)	120	1	
1(b)(ii)	130	1	
1(b)(iii)	60	1	
1(c)(i)	93.4	4	**M1** for mid-values soi **M1** for Σfx **M1 dep on second M** for $\Sigma fx \div 200$
1(c)(ii)	19	2	**M1** for $\frac{86}{50}$ or $\frac{114}{60}$

Question	Answer	Marks	Partial Marks
2(a)	42	2	**M1** for $12 \div 2$ or better
2(b)(i)	5.72	2	**M1** for $\frac{100-12}{100} \times 6.50$ oe or **B1** for 0.88 oe
2(b)(ii)	12.5[0]	2	**M1** for $\frac{100-12}{100} \times x = 11$ or better oe

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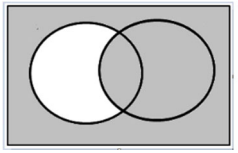
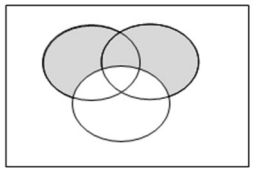
Question	Answer	Marks	Partial Marks
3(c)(i)	[p =] $\frac{1}{2}$ oe [q =] 1	**2**	**B1** for each If zero scored, **SC1** for 2 values satisfying one of the original equations
3(c)(ii)	[u =] 30 and 150 [v =] 0 and 360	**4**	**B1** for each OR **SC1** for sin u = *their p* and cos v = *their q* **SC1** if their two different angles for u sum to 180 or if their different two angles for v sum to 360

Question	Answer	Marks	Partial Marks
4(a)(i)	3	**1**	
4(a)(ii)	7	**1**	**FT** *their* (i) 3×*their* (i) –2
4(b)	$\frac{x+2}{3}$ oe final answer	**2**	**M1** for $y + 2 = 3x$ or $\frac{y}{3} = x - \frac{2}{3}$ or $x = 3y - 2$
4(c)	25	**2**	**M1** for $\frac{1}{x} = 5^{-2}$ oe
4(d)	$\frac{2x^2 - x - 1}{x}$ final answer	**2**	**M1** for $2x - 1 - \frac{1}{x}$
4(e)	2.98×10^{17} or $2.980... \times 10^{17}$	**1**	
4(f)	625	**2**	**M1** for x = j(4)

Question	Answer	Marks	Partial Marks
5(b)(i)	1.27 or 1.272 to 1.273	**2**	**M1** for $\left[\frac{1}{2}\times\right]\pi\times 0.45^2\times 4$ or $\frac{1}{2}\times\pi\times 0.45^2\left[\times 4\right]$
5(b)(ii)	742 or 743	**6**	**M5** for a method leading to the volume of water e.g. $4\times\{2\times\frac{invcos\left(\frac{0.15}{0.45}\right)}{360}\times\pi\times 0.45^2$ $-\frac{1}{2}\times 0.45^2\times\sin\left(2\,invcos\left(\frac{0.15}{0.45}\right)\right)\}$ oe **OR** **M2** $\left[2\times\right]\frac{invcos\left(\frac{0.15}{0.45}\right)}{360}\times\pi\times 0.45^2$ oe or $\left[2\times\right]\frac{90-invcos\left(\frac{0.15}{0.45}\right)}{360}\times\pi\times 0.45^2$ oe or **M1** for use of $\frac{\theta}{360}\times\pi\times 0.45^2$ oe **M2** for $\frac{1}{2}\times 0.45^2\times\sin\left(2\,invcos\left(\frac{0.15}{0.45}\right)\right)$ oe or $\frac{1}{2}\times 0.15\times 0.45\times\sin\left(invcos\left(\frac{0.15}{0.45}\right)\right)\left[\times 2\right]$ oe

Question	Answer	Marks	Partial Marks
5(b)(ii)			**or M1 for use of** $\frac{1}{2}\times 0.45^2$ **× sinθ oe** **or** $[2\times]\frac{1}{2}\times 0.15\times 0.45\times \sin\beta$ **oe** **If 0 scored,** **SC1 for *inv*cos** $\left(\frac{0.15}{0.45}\right)$ **or *inv*sin** $\left(\frac{0.15}{0.45}\right)$ **or** $\sqrt{0.45^2-0.15^2}$ **soi**
6(a)(i)	–3	**1**	
6(a)(ii)	–1 1.55 to 1.6 4.4 to 4.45	**3**	**B1** for each
6(a)(iii)	–8	**1**	
6(a)(iv)	Ruled line through origin intersecting curve once	**2**	**B1** for ruled line through origin
6(b)(i)	18	**3**	**B2** for $6x - 12$ or **B1** for $6x$ or –12
6(b)(ii)	(2, –5)	**2**	**B1** for each. If 0 scored, **M1** for *their* $6x - 12 = 0$ or states $\frac{dy}{dx} = 0$
6(c)	[p =] 7 [q =] 3	**2**	**B1** for each

Question	Answer	Marks	Partial Marks
8(a)(i)	$A \cap B$	**1**	
8(a)(ii)	 	**2**	**B1** for each
8(b)(i)	$\frac{9}{11}$	**1**	
8(b)(ii)	$\frac{36}{121}$ oe	**3**	**M2** for $2 \times \frac{2}{11} \times \frac{9}{11}$ oe or **M1** for $\frac{2}{11} \times \frac{9}{11}$ oe If 0 scored **SC1** for $\frac{36}{110}$
8(c)(i)	3, 5, 28, 14 correctly placed	**2**	**B1** for 28 in the intersection
8(c)(ii)	$\frac{28}{50}$ oe	**1**	**FT** *their* 28 where *their* $28 < 50$
8(c)(iii)	$\frac{123}{175}$ oe	**2**	**M1** for $\frac{42}{50} \times \frac{41}{49}$
8(c)(iv)	$\frac{63}{88}$ oe	**2**	FT *their* 28 **M1** for $\frac{\textit{their}28}{33} \times \frac{\textit{their}28-1}{32}$